

- 3 (a)** weight = $m \times g$
 $= 130.5 \times 9.81 = 1280 \text{ N}$ A1 [1]
- (b) (i)** $F = ma$
 $T - 1280 = 130.5 \times 0.57$ C1
 $T = 1280 + 74.4 = 1350 \text{ N}$ A1 [2]
- (ii)** 1280 N A1 [1]
- (c)** $1240 - 1280 = 130.5 \times a$ C1
 $a = (-) 0.31 \text{ ms}^{-2}$ A1 [2]
- (d) (i)** 1. 3.5 s A1 [1]
2. 6.5 s A1 [1]

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(ii) basic shape M1
correct points A1 [2]

4 (c) force is proportional to extension B1 [1]